

PROJECTIVE MODELS OF HYPERKÄHLER MANIFOLDS AND COBLE TYPE HYPERSURFACES

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ABSTRACT. Hyperkähler manifolds are natural generalizations of K3 surfaces in higher dimensions. Geometric descriptions for locally complete families of projective hyperkähler manifolds are only known in very few cases. In this talk, we first describe the projective geometry of the Hilbert square of a K3 surface of genus 7 or 8, by making use of the Mukai model: in both cases, it can be realized as a degeneracy locus on an ambient homogeneous space. From this, we deduce a geometric description for the two locally complete families of K3^[2]-type (square 4 and square 6 with divisibility 1), in terms of Coble type hypersurfaces. Based on a joint work with Ángel Ríos Ortiz and Andrés Rojas, and an ongoing work also with Benedetta Piroddi.

PART I PROJECTIVE HYPERKÄHLER MANIFOLDS AND WHERE TO FIND THEM

1. K3 SURFACES

Definition 1.1. A *K3 surface* is a compact complex surface S with trivial canonical bundle $\omega_S \simeq \mathcal{O}_S$ and regular (that is, $h^1(\mathcal{O}_S) = 0$).

Having trivial canonical bundle also means that S admits a unique symplectic form: $H^0(S, \Omega_S^2) = \mathbb{C}\omega$.

We will mostly be interested in projective K3 surfaces, that is, those that can be defined using polynomial equations in projective spaces. Let (S, L) be a projective K3 surface with L very ample, so we have an embedding

$$\varphi_L: S \hookrightarrow \mathbb{P}^g.$$

The number g is called the *genus* of S , since a general curve C in the linear system $|L|$ is a canonical curve of genus g . We have $\deg_L S = 2g - 2$.

Examples of projective K3 surfaces. Mukai has described explicitly the projective geometry of a general K3 surface in low degrees through a series of work. These are now known as the *Mukai models*.

Theorem 1.2. *Let (S, L) be a general K3 surface of genus g .*

- $g = 2$: $\varphi_L: S \xrightarrow{2:1} \mathbb{P}^2$ is a double cover branched along a smooth sextic.¹

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¹Note that L is not very ample. Still, this case fits nicely into the picture since genus 2 curves are hyperelliptic.

- $g \in \{3, 4, 5\}$: S is a (4) , $(2, 3)$, $(2, 2, 2)$ -complete intersection in $\mathbf{P}^3$, $\mathbf{P}^4$, $\mathbf{P}^5$, respectively.
- $g = 6$: S is a $(1, 1, 1, 2)$ -complete intersection in $\text{Gr}(2, 5) \subset \mathbf{P}^9$.
- $g = 7$: S is a $\mathbf{P}^7$ -linear section of $\text{OGr}_+(5, 10) \subset \mathbf{P}^{15}$.
- $g = 8$: S is a $\mathbf{P}^8$ -linear section of $\text{Gr}(2, 6) \subset \mathbf{P}^{14}$.
- $g = 9$: S is a $\mathbf{P}^9$ -linear section of $\text{IGr}(3, 6) \subset \mathbf{P}^{13}$.
- $g = 10$: S is a $\mathbf{P}^{10}$ -linear section of $G_2 \subset \mathbf{P}^{13}$.
- $g \in \{11, 12, 13, 16, 18, 20\}$: explicit descriptions.

We note that the deformation theory and the moduli spaces of K3 surfaces are very well understood: for all $g \geq 2$, there exists a (coarse) moduli space $\mathcal{F}_g$ parametrizing K3 surfaces of genus g . This is an irreducible quasi-projective variety of dimension 19. For all the above Mukai models, we indeed obtain a 19-dimensional family,² thus recovering general members in $\mathcal{F}_g$. This is referred to as a *locally complete family* of K3 surfaces.

2. HYPERKÄHLER MANIFOLDS

Hyperkähler manifolds are the natural generalization of K3 surfaces in higher dimensions.

Definition 2.1. A compact Kähler manifold X is called *hyperkähler* if it is simply connected (thus $h^1(\mathcal{O}_X) = 0$) and admits a unique symplectic form: $H^0(X, \Omega_X^2) = \mathbf{C}\omega$.

Note in particular that the existence of a symplectic form ω implies that X must be of even dimension $2n$ and has trivial canonical bundle $\omega_X \simeq \mathcal{O}_X$, since $\omega^{\wedge n}$ provides a trivialization of $\omega_X = \Omega_X^{2n}$.

Again, the deformation theory and the moduli spaces for hyperkähler manifolds are well understood: almost everything is encoded in the second cohomology group $H^2(X, \mathbf{Z})$, which is equipped with a quadratic form q called the *Beauville–Bogomolov–Fujiki form*.

Example 2.2. Following is the list of known deformation types of hyperkähler manifolds in dimension ≥ 4 .

- For a K3 surface S , its Hilbert scheme of n points, denoted by $S^{[n]}$, is a hyperkähler manifold of dimension $2n$. We have

$$H^2(S^{[n]}, \mathbf{Z}) = H^2(S, \mathbf{Z}) \oplus \mathbf{Z}\delta,$$

$$\text{NS}(S^{[n]}) = \text{NS}(S) \oplus \mathbf{Z}\delta,$$

where 2δ is the class of the divisor Δ parametrizing non-reduced subschemes.

Since $b_2(S^{[n]}) = b_2(S) + 1$, general deformations of $S^{[n]}$ are no longer Hilbert schemes. We call them *hyperkähler manifolds of K3^[n]-type*.

- A similar construction for abelian surfaces yields *generalized Kummer varieties* and *hyperkähler manifolds of Kum_n-type*.
- Two sporadic examples constructed by O’Grady in dimension 6 and 10.

Projective models. For K3 surfaces (S, L) , we used the genus g , or equivalently, the degree $\deg_L S = 2g - 2$ to characterize the polarizations. In higher dimensions, however, there are at least two invariants for a polarization H on a hyperkähler manifold X .

- The square $q(H) \in 2\mathbf{Z}$;

²For example, for quartic surfaces, the parameter space is $\mathbf{P}(H^0(\mathbf{P}^3, \mathcal{O}(4))) // \text{SL}(4)$ which has dimension $(35 - 1) - 15 = 19$. It is a fun exercise to obtain the number 19 for the other Mukai models.

- The *divisibility* $\text{div}(H)$ which is the generator for the subgroup $q(H, H^2(X, \mathbf{Z})) \triangleleft \mathbf{Z}$.

From now on we focus on hyperkähler fourfolds of $\text{K3}^{[2]}$ -type. We denote by $\mathcal{M}_{2d}^{(\gamma)}$ the moduli space of polarized (X, H) with $q(H) = 2d$ and $\text{div}(H) = \gamma$.

Proposition 2.3. $\mathcal{M}_{2d}^{(\gamma)}$ is non-empty in the following cases

- Either $\gamma = 1$ and $2d > 0$ is any even integer;
- Or $\gamma = 2$ and $2d \equiv 6 \pmod{8}$.

In all cases, it is always irreducible of dimension 20.

One would like to describe the general members in each locally complete family, in the same vein as the Mukai models for K3 surfaces.

Example 2.4. For $\text{K3}^{[2]}$ -type, only the following examples are known.³

($\gamma = 1$)

- $\mathcal{M}_2^{(1)}$: (O’Grady, 2006) H in this case is not very ample; $X \xrightarrow{2:1} Y_6 \subset \mathbf{P}^5$ are double covers of certain special sextics so they are called *double EPW sextics*. These can be thought of as analogues of K3 surfaces of genus 2.

($\gamma = 2$)

- $\mathcal{M}_6^{(2)}$: (Beauville–Donagi, 1985) $X \subset \text{Gr}(2, 6) \subset \mathbf{P}^{14}$ is the Fano variety of lines $F_1(Y)$ of a smooth cubic fourfold $Y \subset \mathbf{P}^5$. This is the first example to be studied and the most famous / most well-understood case.
- $\mathcal{M}_{22}^{(2)}$: (Debarre–Voisin, 2010) $X \subset \text{Gr}(6, 10) \cap \mathbf{P}^{90}$ is known as the *Debarre–Voisin fourfold*, constructed from an element $\sigma \in \bigwedge^3 V_{10}^\vee$.
- $\mathcal{M}_{38}^{(2)}$: (Iliev–Ranestad, 2001) $X \subset \mathbf{P}^{230}$ is the *variety of sum of powers* of a cubic fourfold Y .

Note in particular that nothing is known for the lowest degree ones $\mathcal{M}_4^{(1)}$ and $\mathcal{M}_6^{(1)}$ for divisibility 1.

3. COBLE HYPERSURFACES

Coble cubics. Before we study the projective models of $\mathcal{M}_4^{(1)}$ and $\mathcal{M}_6^{(1)}$, we first make a digression and review some classical results on projective models of abelian surfaces, involving notably the Coble cubic hypersurfaces.

Consider (A, Θ) a principally polarized abelian surface. For the moment, we assume that A is indecomposable, that is, it is not the product of two elliptic curves. Hence A can be considered as the Jacobian of a smooth curve of genus 2.

The Riemann–Roch formula (as well as Kodaira vanishing) tells us that $h^0(A, k\Theta) = k^2$ for $k \geq 1$. When $k = 2$, 2Θ is basepoint-free and we get a morphism

$$\varphi_{2\Theta}: A \longrightarrow \mathbf{P}^3.$$

This is however not an embedding: it is two-to-one, and the image is precisely the Kummer surface $K(A) := A/\{\pm 1\}$, a singular quartic surface with sixteen nodes. When $k = 3$, we get a morphism

$$\varphi_{3\Theta}: A \hookrightarrow \mathbf{P}^8$$

³For completeness, there are two more families known in higher dimensions: one of $\text{K3}^{[4]}$ -type ${}^4\mathcal{M}_2^{(2)}$ by Lehn–Lehn–Sorger–van Straten in 2017, and one of $\text{K3}^{[3]}$ -type ${}^3\mathcal{M}_4^{(2)}$ by Iliev–Kapustka–Kapustka–Ranestad in 2019.

which is this time an embedding. Moreover, the geometry of A in $\mathbf{P}^8$ can be described by the following.

Theorem 3.1 (Coble, 1917). *The image of A in $\mathbf{P}^8$ is contained in 9 quadrics $\frac{\partial f_3}{\partial x_i}$ that are the polars of a cubic hypersurface $\mathcal{C}_3 := V(f_3)$. In other words, the singular locus $\text{Sing}(\mathcal{C}_3)$ contains A .*

It is not until much later that we have the following.

Theorem 3.2 (Barth, 1995). *The image of A in $\mathbf{P}^8$ is scheme-theoretically cut out by the 9 quadrics. In other words, the singular locus $\text{Sing}(\mathcal{C}_3)$ is exactly A .*

Degeneration. In the literature, the existence of the Coble cubic is usually either proved via explicit construction of equations, or via abstract group representation theory (see also Section 8). We now present a way to conceptually see why such a cubic should exist.

We consider the “degenerated” case, where the abelian surface A becomes decomposable, that is, A can be written as the product $E_1 \times E_2$ of two elliptic curves. The polarization 3Θ now becomes a $(3, 3)$ -divisor: for each component E_i , we consider the complete linear system that embeds E_i in a $\mathbf{P}^2$ as a cubic curve. Therefore, the map from A to $\mathbf{P}^8$ factorizes through the Segre embedding $\mathbf{P}^2 \times \mathbf{P}^2 \hookrightarrow \mathbf{P}^8$. We have the following picture

$$\begin{array}{ccccc} & & & Y & \\ & & \nearrow & & \searrow \\ A = E_1 \times E_2 & \hookrightarrow & \mathbf{P}^2 \times \mathbf{P}^2 & \xrightarrow{\text{Segre}} & \mathbf{P}^8. \end{array}$$

Note that in this picture, we have a natural cubic hypersurface Y in $\mathbf{P}^8$ whose singular locus is $\mathbf{P}^2 \times \mathbf{P}^2$. Namely, the projective space $\mathbf{P}^8$ can be viewed as the projective space of 3×3 matrices. And Y and $\mathbf{P}^2 \times \mathbf{P}^2$ naturally appear as the rank loci defined respectively by the vanishing of the determinant, which is a cubic polynomial, and the vanishing of the nine 2-minors, which are quadrics and are precisely the partial derivatives of the determinant.

$$\begin{aligned} \mathbf{P}^8 = \mathbf{P}M_{3 \times 3} &= \left\{ \text{nonzero } 3 \times 3 \text{ matrix } M \text{ of the form } \begin{pmatrix} x_0 & x_1 & x_2 \\ x_3 & x_4 & x_5 \\ x_6 & x_7 & x_8 \end{pmatrix} \right\} / \mathbf{C}^* \\ &\cup \\ Y &= \{M \mid \text{rank } M \leq 2\} / \mathbf{C}^* = V(\det) \\ &\cup \\ \mathbf{P}^2 \times \mathbf{P}^2 &= \{M \mid \text{rank } M \leq 1\} / \mathbf{C}^* = V(2\text{-minors}) = V\left(\frac{\partial \det}{\partial x_i}\right) \end{aligned}$$

We thus obtain a highly degenerated “Coble cubic” Y whose singular locus is the fourfold $\mathbf{P}^2 \times \mathbf{P}^2$, which contains all images of decomposable abelian surfaces $E_1 \times E_2$. To recover the abelian surface, we note that it can be seen as the singular locus of another cubic: this is the reducible hypersurface $\Sigma \subset \mathbf{P}^2 \times \mathbf{P}^2$ consisting of the two components $E_1 \times \mathbf{P}^2$ and $\mathbf{P}^2 \times E_2$, and its singular locus is precisely the intersection of the two components, namely, $E_1 \times E_2$.

Thus we have the following picture

$$\begin{array}{ccc}
 (E_1 \times E_2, (3, 3)) & \leftarrow \text{~~~~~} & (A, 3\Theta) \\
 \text{second cubic } \Sigma \downarrow & & \parallel \\
 \mathbf{P}^2 \times \mathbf{P}^2 = \text{Sing}(Y) & \leftarrow \text{~~~~~} & \text{Sing}(\mathcal{C}_3).
 \end{array}$$

4. HYPERKÄHLER FOURFOLDS AND COBLE TYPE HYPERSURFACES

We now go back to hyperkähler manifolds and announce our main results.

Theorem 4.1. *We have the following descriptions for the two locally complete families.*

$(\mathcal{M}_4^{(1)})$

- For a general K3 surface S of genus 7, the projective hyperkähler fourfold $(S^{[2]}, L - 2\delta)$ (with $q(L - 2\delta) = 4$) is embedded in $\mathbf{P}^9$.

The image of $S^{[2]}$ is contained in a quadric Q (the singular locus $\text{Sing}(V(q^2))$ of the non-reduced “double quadric” which is a quartic), as the singular locus of a unique quartic $D \subset Q$.

- A general $(X, H) \in \mathcal{M}_4^{(1)}$ is embedded in $\mathbf{P}^9$. There exists a unique quartic $Y = V(f_4)$ such that $\text{Sing}(Y) = X$.

$(\mathcal{M}_6^{(1)})$

- For a general K3 surface S of genus 8, the projective hyperkähler fourfold $(S^{[2]}, L - 2\delta)$ (with $q(L - 2\delta) = 6$) is embedded in $\mathbf{P}^{14}$.

The image of $S^{[2]}$ is contained in a Grassmannian $\text{Gr}(2, 6)$ (the singular locus $\text{Sing}(V(\text{pf}))$ of the Pfaffian cubic), as the singular locus of a unique cubic $D \subset \text{Gr}(2, 6)$.

- A general $(X, H) \in \mathcal{M}_6^{(1)}$ is embedded in $\mathbf{P}^{14}$. There exists a unique cubic $Y = V(f_3)$ such that $\text{Sing}(Y) = X$.

Observe that in both cases, we have a picture remarkably similar to the classical Coble cubics, where the general Coble type hypersurface is singular *exactly* along our variety (A or X), while a particular degeneration gives us a hypersurface with a larger singular locus ($\mathbf{P}^2 \times \mathbf{P}^2$ or Q or $\text{Gr}(2, 6)$) that contains all images of the “special” members (decomposable abelian surfaces or Hilbert squares of K3), and a second hypersurface of the same degree (Σ or D) is required to determine these members in this larger singular locus.

Of course, this is not the end of the story. There still remains the question of parametrizing these hypersurfaces, so that we can write down the equations of a general member, in the same way as for the other locally complete families.

PART II

5. LINEAR SYSTEMS ON $S^{[2]}$

As we do not have the description of a general member, the best we can do is to study Hilbert squares $S^{[2]}$ and try to deduce from them properties that a general member may satisfy.

Let $(S, L) \subset \mathbf{P}^g$ be a projective K3 surface of genus g . Let us first consider the polarization $L - \delta$ on $S^{[2]}$: it is given by the following natural map

$$(1) \quad \begin{aligned} \varphi_{L-\delta}: S^{[2]} &\longrightarrow \mathrm{Gr}(2, g+1) \\ (x, y) &\longmapsto [\overline{xy}] \end{aligned}$$

where a subscheme is sent to the class of the line it spans. We have $\varphi_{L-\delta}^* \mathcal{O}(1) = L - \delta$. Note that $L - \delta$ is very ample if and only if L is *2-very ample* on S , by definition. We have the following result.

Proposition 5.1. *Let (S, L) be a K3 surface of genus g . Assuming moreover that $\mathrm{NS}(S) = \mathbf{Z}L$. Then*

- *When $g \geq 5$, L is 2-very ample.*
- *When $g \geq 3$, $2L$ is 2-very ample.*

Since being very ample is an open property, we get the following.

Corollary 5.2. *A general $(X, H) \in \mathcal{M}_{2d}^{(\gamma)}$ is very ample when $2d \geq 6$, for either divisibility 1 or 2.*

Thus for very ampleness, $\mathcal{M}_4^{(1)}$ remains the last unknown case.

Now we consider the polarization $L - 2\delta$. We first recall a classical result concerning the ideal of S in $\mathbf{P}^g$.

Theorem 5.3 (Saint-Donat). *Let S be a projective K3 surface.*

- *S is projectively normal.*
- *When $g \geq 5$ and S is general, the ideal of S is generated by quadrics.*

Let (S, L) be a general K3 surface of genus $g \geq 5$ with $\mathrm{NS}(S) = \mathbf{Z}L$ so that the ideal of S is generated by quadrics. We consider the following map

$$(2) \quad \begin{aligned} \varphi_{L-2\delta}: S^{[2]} &\longrightarrow |I_S(2)|^\vee \\ (x, y) &\longmapsto \{Q \in I_S(2) \mid Q \supset \overline{xy}\} \end{aligned}$$

where a subscheme is mapped to the hyperplane of quadrics containing the line it spans; since S cannot contain the line by the assumption that $\mathrm{NS}(S) = \mathbf{Z}L$, the map is indeed everywhere defined. We have $\varphi_{L-2\delta}^* \mathcal{O}(1) = L - 2\delta$.

Our first contribution is the following.

Theorem 5.4. *Let (S, L) be a K3 surface of genus $g \geq 7$ with $\mathrm{NS}(S) = \mathbf{Z}L$. Then $(S^{[2]}, L - 2\delta)$ is very ample.⁴*

By taking $g = 7$ this settles very ampleness for the remaining case $\mathcal{M}_4^{(1)}$. This also ensures us that in the two cases $g \in \{7, 8\}$ we are considering, the Hilbert square $S^{[2]}$ is indeed embedded in the projective space via the linear system.

⁴The proof involves a detailed analysis of the rank loci in $|I_S(2)|$, and their relations with moduli spaces of stable sheaves on S .

6. PROJECTIVE GEOMETRY OF $(S^{[2]}, L - 2\delta)$

The goal is now to understand the projective geometry of $(S^{[2]}, L - 2\delta)$, in the two cases of $g \in \{7, 8\}$ (they are elements of $\mathcal{M}_4^{(1)}$ and $\mathcal{M}_6^{(1)}$, respectively), by making use of their Mukai models. Recall that

- When $g = 7$, a general S is a $\mathbf{P}^7$ -linear section of $\mathrm{OGr}_+(5, 10) \subset \mathbf{P}^{15}$;
- When $g = 8$, a general S is a $\mathbf{P}^8$ -linear section of $\mathrm{Gr}(2, 6) \subset \mathbf{P}^{14}$.

We focus on the case of $g = 7$; the case of $g = 8$ turns out to be extremely similar.

Orthogonal Grassmannian. First, we recall some basic properties of the orthogonal Grassmannian $\mathrm{OGr}_+(5, 10)$. Let V_{10} be a 10-dimensional vector space equipped with a quadratic form $q \in \mathrm{Sym}^2 V_{10}^\vee$. Note that the quadratic form provides us with an isomorphism $V_{10} \simeq V_{10}^\vee$. The orthogonal Grassmannian $\mathrm{OGr}_+ := \mathrm{OGr}_+(5, V_{10})$ is one of the two components parametrizing 5-dimensional (maximal) isotropic subspaces with respect to q . It has dimension 10 and admits the *spinor embedding*

$$\mathrm{OGr}_+ \hookrightarrow \mathbf{P}^{15} = \mathbf{P}V_+,$$

where V_+ is one of the two spinor representations of $\mathrm{SO}(10)$ of rank 16. We remark that twice the spinor embedding (that is, taking the Veronese of degree 2) gives us the Plücker embedding.

Proposition 6.1. *The ideal of OGr_+ is generated by quadrics, and there is a canonical identification $I_{\mathrm{OGr}_+}(2) = V_{10}$.*

Since S is a linear section of OGr_+ , in consequence, we get also the identification $I_S(2) = I_{\mathrm{OGr}_+}(2) = V_{10}$.

Lemma 6.2. *For $x, y \in S$, let U_x and U_y be the corresponding 5-dimensional isotropic subspaces. We have $\dim(U_x \cap U_y) = 1$, $\dim(U_x + U_y) = 9$.*

Thus we get two ways of mapping $S^{[2]}$ into a projective space.

Proposition 6.3.

- *The map*

$$\begin{aligned} \varphi: S^{[2]} &\longrightarrow \mathbf{P}V_{10}^\vee = |I_S(2)|^\vee \\ (x, y) &\longmapsto U_x + U_y \end{aligned}$$

coincides with the map $\varphi_{L-2\delta}$ in (2).⁵

- *Composing φ with the isomorphism $\mathbf{P}V_{10}^\vee \simeq \mathbf{P}V_{10}$ induced by the quadratic form q , we get*

$$\begin{aligned} \psi: S^{[2]} &\longrightarrow \mathbf{P}V_{10} = |I_S(2)| \\ (x, y) &\longmapsto U_x \cap U_y. \end{aligned}$$

The quadric $Q_{x,y}$ in the image is the unique quadric singular along the line $\overline{xy}$.

A first consequence is that, via the embedding ψ , $S^{[2]}$ is contained in the quadric $Q \subset \mathbf{P}V_{10}$, as $U_x \cap U_y$ must still be isotropic. Note that we have

$$h^0(\mathbf{P}^9, \mathcal{O}(2)) = 55 = h^0(S^{[2]}, \mathcal{O}(2)),$$

so $S^{[2]}$ lying on a quadric in particular means it is *not* projectively normal.

⁵For the proof, we check that, under the identification $V_{10} = I_S(2)$, U_x and U_y consist of quadrics containing the line $\overline{xy}$.

Degeneracy locus. Now we present a nice geometric construction to realize $S^{[2]}$ as a degeneracy locus in Q .

Consider the following incidence variety

$$\begin{array}{ccccc}
 & I_p & I := \{(p, [U]) \mid p \in \mathbf{P}(U)\} & & \\
 & \swarrow & \searrow \text{pr}_1 & \searrow \text{pr}_2 & \\
 p & & \mathbf{P}^9 \supset Q & & \text{OGr}_+ \subset \mathbf{P}^{15} = \mathbf{P}V_+
 \end{array}$$

For each $p \in Q$, the preimage $I_p := \text{pr}_1^{-1}(p)$ is a subvariety of OGr_+ of dimension 6. It is a classical result that I_p spans a $\mathbf{P}^7$ in $\mathbf{P}^{15}$, inside which it is a quadric hypersurface, and when globalizing this to the entire Q , we obtain an 8-dimensional vector bundle $\mathcal{S}_+$ known as the *spinor bundle*. Clearly, via

$$\mathcal{S}_+ \hookrightarrow V_+ \otimes \mathcal{O}_Q,$$

we can view $\mathcal{S}_+$ as a subbundle of the rank-16 trivial bundle $V_+ \otimes \mathcal{O}_Q$.

On the other hand, (S, L) is a $\mathbf{P}^7$ -linear section of OGr_+ . In other words, the linear span $H^0(S, L)^\vee$ is an 8-dimensional subspace of V_+ so it naturally admits an 8-dimensional quotient W

$$0 \longrightarrow H^0(S, L)^\vee \longrightarrow V_+ \longrightarrow W \longrightarrow 0.$$

Via composition, we get a map between two vector bundles of rank 8

$$\varphi: \mathcal{S}_+ \longrightarrow W \otimes \mathcal{O}_Q.$$

From the construction, it is not hard to deduce the following.

Proposition 6.4. *Via the embedding $\psi: S^{[2]} \hookrightarrow Q \subset \mathbf{P}^9$, $S^{[2]}$ coincides with the degeneracy locus $D_6(\varphi)$ scheme-theoretically.*

Note that the first degeneracy locus $D := D_7(\varphi)$ is a hypersurface in Q defined by the determinant $\det(\varphi)$. In particular, we can check that it is a quartic. By a Bertini-type argument, we obtain the following.

Corollary 6.5. *For a general S of genus 7, the singular locus of the quartic $D = D_7(\varphi)$ is exactly the next degeneracy locus $S^{[2]} = D_6(\varphi)$.*

Since we can realize $S^{[2]}$ as a degeneracy locus, we now have the full machinery of determinantal ideals at our disposal. In the particular case of corank-2 square matrices, there is the four-term locally-free *Gulliksen–Negård* resolution

$$0 \longrightarrow F_4 \longrightarrow F_3 \longrightarrow F_2 \longrightarrow F_1 \longrightarrow \mathcal{I}_{S^{[2]}/Q} \longrightarrow 0.$$

Proposition 6.6. *Using the Gulliksen–Negård complex,⁶ we obtain*

- $S^{[2]}$ is d -normal for $d \geq 3$, and $I_{S^{[2]}}(2) = \mathbf{C}Q$ (so 2-normality only fails by one).
- $\mathcal{I}_{S^{[2]}/Q}$ is generated by 65 quartics.

⁶Each F_i in the complex can be explicitly described in terms of homogeneous vector bundles on Q . Therefore, one can compute various cohomology groups on $S^{[2]}$ using the associated spectral sequence and the Borel–Weil–Bott theorem.

- We can describe the full minimal resolution in terms of its Betti diagram, which is

	0	1	2	3	4	5	6	7
0	1	.	.	.	.	.	.	.
1	.	<u>1</u>	.	.	.	.	.	.
2	<u>1</u>	10	.	.	.	.	.	.
3	.	20	126	190	130	<u>46</u>	10	1
4	.	.	.	.	<u>1</u>	.	.	.
5	.	.	.	.	.	1	.	.

The four terms that are marked with underline contain extra syzygies not expected from computing with the Hilbert function.

The natural question now is what happens with a general deformation of $S^{[2]}$. With some computation of infinitesimal nature (namely, by counting first-order deformations of $S^{[2]}$ in Q and in $\mathbf{P}^9$), we are able to conclude that

Proposition 6.7. *The space of first-order deformations $H^0(S^{[2]}, \mathcal{N}_{S^{[2]}/Q})$ of $S^{[2]}$ in Q can be identified with $H^0(S, L) \otimes W$, the 64-dimensional tangent space at $[H^0(S, L)^\vee]$ to the Grassmannian $\text{Gr}(8, V_+)$.*

In other words, deforming $S^{[2]}$ in Q is essentially the same as deforming the K3 surface S . Thus a general deformation of $S^{[2]}$ in $\mathbf{P}^9$ cannot be contained in a quadric.

Corollary 6.8. *A general deformation of $(S^{[2]}, L - 2\delta)$, that is, a general $(X, H) \in \mathcal{M}_4^{(1)}$, is not contained in any quadrics. Therefore it is projectively normal with Betti diagram*

	0	1	2	3	4	5	6	7
0	1	.	.	.	.	.	.	.
1	.	.	.	.	.	.	.	.
2	.	10	.	.	.	.	.	.
3	.	20	126	190	130	45	10	1
4	.	.	.	.	.	.	.	.
5	.	.	.	.	.	1	.	.

In particular, its homogeneous ideal is generated by 10 cubics with no linear syzygies and 20 quartics.

Once we prove the existence of the Coble type quartic, it becomes clear that the ten cubics are precisely its polars.

7. PROOF OF EXISTENCE OF THE QUARTIC

We now explain the idea of the proof for the existence of the Coble type quartic.

Theorem 7.1. *For a general $(X, H) \in \mathcal{M}_4^{(1)}$ embedded in $\mathbf{P}^9$, there exists a unique quartic $Y = V(f_4)$ such that $\text{Sing}(Y) = X$.*

Proof. First, we prove the existence of a quartic $Y = V(f_4)$ whose singular locus $\text{Sing}(Y)$ contains X . For this, it suffices to show that $h^0(\mathbf{P}^9, \mathcal{I}_X^2(4)) \geq 1$. Note that we have the exact sequence

$$0 \longrightarrow \mathcal{I}_X^2(4) \longrightarrow \mathcal{I}_X(4) \longrightarrow \mathcal{I}_X/\mathcal{I}_X^2(4) = \mathcal{N}_{X/\mathbf{P}^9}^\vee(4) \longrightarrow 0,$$

where we know $h^0(\mathbf{P}^9, \mathcal{I}_X(4)) = 120$ since X is projectively normal. Thus it suffices to show that $h^0(X, \mathcal{N}_{X/\mathbf{P}^9}^\vee(4)) \leq 119$. We can specialize to $S^{[2]}$, compute that $h^0(S^{[2]}, \mathcal{N}_{S^{[2]}/\mathbf{P}^9}^\vee(4)) = 55 + 64 = 119$, and conclude using upper semicontinuity.

Note that since X is contained in exactly 10 cubics, such quartic must be unique.

Second, we need to show that the singular locus of Y is *exactly* X . We consider a general deformation

$$\begin{array}{c} \mathcal{X} \subset \mathbf{P}^9 \times \Delta \\ \downarrow \\ \Delta \end{array}$$

with central fiber $\mathcal{X}_0$ being $(S^{[2]}, L - 2\delta)$ for a general K3 surface S of genus 7. We obtain a family of quartics $\mathcal{Y}$ with $\text{Sing}(\mathcal{Y}_t) \supset \mathcal{X}_t$ and the central fiber $\mathcal{Y}_0$ being the “double quadric” $V(q^2)$. We can then define an intermediate family $\mathcal{Z} \subset \mathcal{Y}$, where $\mathcal{Z}_t$ is exactly $\text{Sing}(\mathcal{Y}_t)$ for $t \neq 0$. We assume that the family $\mathcal{Y}$ is given by

$$\mathcal{Y} = V(F) \text{ where } F = q^2 + t \cdot f + o(t),$$

and check that the “tangential quartic” f restricts to the quartic hypersurface D in Q , that is, $V(q, f) = D$. Then the singular locus of D being $S^{[2]}$ means that we can complete the family $\mathcal{Z}$ with $\mathcal{Z}_0$ being $S^{[2]} = \mathcal{X}_0$. By upper semicontinuity, this forces all $\mathcal{Z}_t$ to coincide with $\mathcal{X}_t$. $\square$

8. PARAMETRIZATION OF $\mathcal{M}_4^{(1)}$?

For the classical Coble cubics $(A, 3\Theta) \subset \mathcal{C}_3 \subset \mathbf{P}^8$, there is a natural way to study the moduli space. Namely, since A comes with a group structure, this makes $\mathbf{P}^8$ into a projective representation of the *Heisenberg group* H , and the Coble cubic $\mathcal{C}_3$ must be H -invariant. Therefore, instead of considering all cubic polynomials on $\mathbf{P}^8$, it suffices to consider just those that are H -invariant, which turns out to be

$$\mathbf{P}(H^0(\mathbf{P}^8, \mathcal{O}(3))^H) = \mathbf{P}^4.$$

And the Coble cubics form the *Burkhardt quartic* $\mathcal{B}_4 \subset \mathbf{P}^4$. In this way, one recovers the 3-dimensional locally complete family of Coble cubics (since $\dim \mathcal{A}_2 = 3$).

In the case of hyperkähler manifolds, we do not have a group action in general. However, we can restrict ourselves to those *with* a group action, and mimic the same trick to try to produce some explicit parametrizations. Indeed, for the Mathieu group M_{20} (of order 960)—the largest possible group G such that there exists a positive dimension family of $(X, H) \in \mathcal{M}_4^{(1)}$ with a symplectic G -action—we can produce an explicit one-dimensional family

$$\mathcal{C}_{M_{20}} \subset \mathbf{P}(H^0(\mathbf{P}^9, \mathcal{O}(4))^{M_{20}}) = \mathbf{P}^4$$

that dominates the one-dimensional family $\mathcal{M}_{4, M_{20}}^{(1)}$ of (X, H) with a symplectic M_{20} -action. In particular, the “double quadric” $[q^2]$ lies on $\mathcal{C}_{M_{20}}$, and we obtain explicitly the “tangential quartic” f that determines the image of $S^{[2]}$ in Q , where S is the unique K3 surface of genus 7 admitting a symplectic M_{20} -action. Moreover, there are several other types of interesting degenerations and special geometric conditions appearing in this family. This could thus be thought of as a baby example towards understanding the whole moduli space $\mathcal{M}_4^{(1)}$.